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When Roads Go Underwater: AI-Enhanced Digital Twin–Driven Flood Resilience for Roadway Serviceability Assessment

Moeid Shariatfar, Ph.D.[1]; Yangtian Yin[2]; Yong-Cheol Lee, Ph.D., A.M.ASCE[3]; Kunhee Choi, Ph.D., M.ASCE[4]

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Abstract: Extreme flooding poses escalating risks to roadway infrastructure, threatening structural integrity, operational reliability, and public safety. Existing flood-monitoring approaches primarily utilize sparse sensor networks, thus providing limited real-time data and insufficiently capturing indirect flooding impacts on noninundated roadway segments. This gap complicates emergency response, delays evacuation, and undermines postevent recovery efforts. Addressing this critical knowledge gap, this study proposes an advanced predictive decision-support framework leveraging an artificial intelligence (AI)-enhanced digital twin integrated with flood simulations and graph neural network (GNN) modeling. It systematically assesses roadway serviceability during extreme flooding by integrating structural conditions, operational disruptions, historical maintenance records, inundation severity, and recovery timelines. By consolidating heterogeneous data sets, including historical traffic volumes, pavement conditions, hydrological data, and weather forecasts, the developed framework provides accurate, real-time predictive insights for both inundated and indirectly impacted roadway segments. This capability was demonstrated through an illustrative validation study. Ultimately, this research equips transportation agencies and emergency responders with robust hands-on tools that can facilitate optimized emergency response, improved infrastructure management, and enhanced resilience of transportation systems under ever-increasing flood risks due to climate change. **DOI: 10.1061/JMENEA.MEENG-7046.** *© 2026 American Society of Civil Engineers.*

Introduction, Gaps in Knowledge, and Significance

Road networks are susceptible to natural disasters such as floods that impact both the structural integrity of the roadways and their long-term serviceability. Considering the short-term implications for road usability is essential because debris from disasters and flooding can lead to road blockages (Diakakis et al. 2020). During these disasters, the increased traffic due to mainline lane closures or people trying to escape the affected area can further affect the road network's serviceability unfavorably (Mitsakis et al. 2014). Typically, following such an event, specific locations may become less accessible, which is especially critical for first responders in some cases and for people who are dispatched for rescue operations and emergent support.

Effective monitoring and assessment of roadway conditions during extreme flooding events are essential to safeguarding public safety, supporting timely emergency response, and enhancing the resilience of transportation infrastructure. Current flood-monitoring approaches predominantly utilize physical sensor systems deployed along rivers and waterways to track rising water levels and potential overflow events. However, these systems offer valuable insights regarding local inundation; their sparse geographical distribution results in incomplete and often fragmented data regarding the broader impact of flooding across extensive roadway networks. Furthermore, these systems do not provide accurate, real-time flooding status updates for individual roadway segments, significantly limiting their effectiveness in critical decision-making scenarios during disasters (Yuan et al. 2022).

Urban and infrastructure planning processes traditionally leverage floodplain mapping, historical flooding data, and established hydrological models to anticipate and mitigate flooding risks. Nevertheless, unprecedented climate phenomena, rapid urbanization, altered land-use patterns, and evolving topographic conditions have introduced substantial uncertainties into these conventional flood-risk assessments. Consequently, the reliability of existing floodplain-based planning and management frameworks has diminished, highlighting the urgent need for dynamic monitoring solutions that integrate real-time and historical data streams to accurately reflect evolving environmental and infrastructural conditions.

A critical operational challenge arising from this data inadequacy is the significant disruption to first responders' capabilities during flood events. The absence of integrated real-time monitoring and predictive tools results in uncertainty regarding roadway accessibility, closures, and congestion, complicating rescue operations, delaying response activities, and increasing safety risks for both responders and affected populations (Helderop and Grubesic 2019). Furthermore, previous research efforts have rarely explored the identification of feasible alternative routes or systematically assessed

[1]Transportation Engineer, Caltrans Division of Engineering Services, P.O. Box 942873, Sacramento, CA 95816-8041. ORCID: https://orcid.org/0000-0001-5272-1406. Email: Moeid.Shariatfar@dot.ca.gov

[2]Ph.D. Candidate, Dept. of Construction Science, Texas A&M Univ., 3137 TAMU, Francis Hall 328, College Station, TX 77843-3137. ORCID: https://orcid.org/0000-0001-5769-374X. Email: yinyangt@tamu.edu

[3]Associate Professor, Bert S. Turner Dept. of Construction Management, Louisiana State Univ., 3315-E Patrick F. Taylor Hall, Baton Rouge, LA 70803 (corresponding author). ORCID: https://orcid.org/0000-0002-0040-0894. Email: yclee@lsu.edu

[4]Professor, Dept. of Construction Science, Texas A&M Univ., 3337 TAMU, Francis Hall 216C, College Station, TX 77843-3137. ORCID: https://orcid.org/0000-0002-0184-2977. Email: kchoi@tamu.edu

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accessibility to essential locations under disrupted transportation scenarios, further exacerbating response-related challenges during catastrophic flooding.

Current methodologies for evaluating roadway serviceability have primarily focused on surface conditions and structural integrity, offering essential yet limited insights into infrastructure performance and maintenance needs (Aleadelat et al. 2018; Fuentes et al. 2021). Despite recent significant advancements in modeling flooding impacts on transportation networks, most existing studies emphasized flood propagation modeling, highlighting inundation severity and spatial extent, but seldom comprehensively addressed the indirect impacts of flooding on roadway segments not directly submerged but still adversely influenced by nearby inundation at a network level (Yuan et al. 2022). As a result, prevailing assessment practices significantly underestimate the true scope of network disruption and fail to capture the complexity of roadway serviceability restoration efforts following severe flooding events.

Considering the existing challenges in the current methodologies, this study is guided by the following research question: How can a predictive, segment-level assessment of roadway serviceability under flood conditions be achieved in near real time by integrating traffic patterns, flood simulation, and roadway attributes? This question drives the development and validation of an integrated framework that goes beyond binary inundation mapping to forecast both direct and indirect service impacts across a dynamic road network.

To bridge this critical research gap, a broader and more holistic analytical approach is required to accurately define and evaluate roadway serviceability under flooding conditions at a macroscopic system network level. Specifically, this study defines serviceability as a multifaceted measure that integrates operational capacity, inundation severity, and flood duration. Leveraging extensive historical data sets, including pavement performance records, traffic flow data, infrastructure conditions, and historical flooding events, this research proposes an advanced, network-level evaluation framework. This methodology revolves around the creation and implementation of a digital twin framework seamlessly integrated with flood simulation capabilities and graph neural network (GNN) models. This innovative integration approach facilitates holistic, real-time simulations, precise visualizations, and predictive analyses of flooding impacts across entire roadway networks. These analyses encompass both directly inundated and indirectly affected roadway segments.

In this study, we adopt the concept of a digital twin (DT) as a dynamic, virtual replica of a physical system that integrates real-time data streams, supports predictive analytics, and provides an interactive interface for operational decision-making. Unlike conventional simulation models, which often operate in isolation or with static inputs, digital twins are characterized by their ability to receive continuous updates from physical assets and simulate system behavior under evolving conditions. Although some digital twin implementations also include bidirectional control (e.g., sending actions back to physical systems), our focus lies on enabling real-time monitoring and forecasting based on live and historical data, within a responsive GIS-based environment. This qualifies the current system as an operational digital twin designed for flood event response.

By addressing these fundamental limitations and questions in existing research and operational practice, this study aims to provide transportation agencies and emergency responders with robust, actionable insights for improved decision-making during extreme flooding scenarios. Ultimately, the anticipated outcomes include better-informed evacuation strategies, optimized emergency response, more effective maintenance prioritization, and significantly enhanced resilience, safety, and operational reliability of transportation infrastructure in the face of increasing flooding risks.

Research Objectives and Methods

The overarching goal of this study is to address the substantial knowledge gap regarding the intricate relationships among roadway serviceability, historical traffic data, and roadway performance under extreme flooding scenarios. Specifically, this study seeks to develop an AI-enhanced digital twin–driven predictive decision-support framework that systematically and efficiently assesses roadway conditions during severe flooding by leveraging traffic operation data to establish baseline performance metrics for forecasting future roadway serviceability.

Building upon prior research that successfully simulated flooding impacts on directly inundated road segments and their accessibility to critical locations, this study expands the analytical scope to holistically evaluate broader network-level implications, including roadway segments indirectly affected by flooding events. To achieve this goal, the research developed and implemented an integrated methodology capable of assessing roadway serviceability from a short-term perspective, focusing on immediate impacts such as accessibility, operational disruption, and traffic-flow disruptions.

A key objective of this research involves the construction and validation of an advanced digital twin framework. This innovative approach integrates flood simulation and state-of-the-art GNN modeling techniques to dynamically simulate, visualize, and analyze the impact of flooding across entire transportation networks. The digital twin developed for this study assessed inundation severity, flood duration, traffic conditions, structural conditions, and historical pavement performance data, providing accurate predictive diagnostics into both inundated and indirectly impacted roadway segments.

Furthermore, this study also aims to create a robust visualization and predictive decision-support tool capable of consolidating diverse and heterogeneous data sets, including weather forecasts, hydrological information, historical traffic volumes, flood event records, and pavement condition metrics. This integrated platform can deliver real-time situational awareness and predictive analytics to facilitate more effective evacuation planning, optimize emergency response operations, and improve strategic decision-making processes related to disaster resilience and infrastructure management.

In summary, the objective of this study is to develop a real-time decision-support framework that enhances flood resilience in roadway networks. Specifically, this study aims to achieve the following objectives:

- Define and quantify roadway serviceability under flood conditions as a function of inundation status, and operational performance.
- Develop a predictive model using a graph attention network (GAT) that incorporates flood simulation data and historical traffic patterns to estimate short-term speed and accessibility across the network.
- Integrate the predictive model within a digital twin visualization environment for interactive, GIS-based decision support during flood events.
- Validate the proposed framework using historical flood and traffic data from Hurricane Harvey to demonstrate its capability in capturing both direct and indirect roadway impacts.

Literature Review

The quality of road service can be defined as the capability of either a product or a service to satisfy the potential or actual needs of the users or, more in general, of the beneficiaries of the product/service itself (Colonna et al. 2004). There is a vulnerability factor for assessing the

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serviceability of a roadway section. A vulnerability factor is a score that measures how sensitive a roadway is to specific risks, like floods. It is used to assess the potential for an event with consequences caused by hazards. For roads, flood vulnerability is influenced by location, structure, elevation, and drainage. There are two aspects of flooding vulnerability for a roadway section (Tao and Mallick 2020):

- Vulnerability factor 1 (VF1) reflects the chance of a roadway section being flooded during a storm. It is determined by the road's location, elevation, and hydrological conditions.
- Vulnerability factor 2 (VF2) indicates how likely a flooded roadway section's structural capacity will deteriorate due to flooding. It depends on factors like water infiltration, structural condition, subgrade sensitivity, and drainage (Shariatfar et al. 2022; Lee et al. 2020; Shariatfar and Lee 2024).
- The overall vulnerability factor (VF) is the product of VF1 and VF2.

This study aims to provide a holistic roadway serviceability assessment framework that considers short-term roadway service in areas susceptible to flooding. The introduced framework incorporates VF1 to analyze and predict short-term flood-induced vulnerability, enabling first responders to identify the most serviceable routes for reaching critical locations during emergency events. It predicts accessible and inaccessible roadways in case of flooding and their effect on the entire roadway network. Carey and Irick (1960) described relative performance as how much pavement can serve its customer during a service period, although it could differ from everyone's perspective. Serviceability of a specific section of a pavement is defined as the ability to serve the existing condition in high-speed, high-volume, and mixed traffic (truck and automobile) situations. The definition is applied to primary highways and road tests and should be modified for use on city streets, farm roads, and so on.

Vulnerability defines how much transportation assets are susceptive to harsh weather and climate effects that are exposed to them (Filosa et al. 2017). On the other hand, the capability of coping with the existing climate variability or future climate situations is defined as adaptive capacity, which is essential to know the weather variables to evaluate the level of risk in different climate conditions. Also, the Federal Transit Administration considers vulnerability assessment an evaluation process to understand how natural hazards may endanger public transport systems. Using six resilience phases of Assess, Plan, Design and Construct, Manage, Maintain, and Monitor, the *Guidebook* provides a resilience planning framework to address vulnerabilities and build resilience to extreme weather conditions.

Short-Term Effect on Serviceability

Many researchers suggest that there is no single definition suitable for vulnerability, but it must be defined in the context of an event. A definition that describes vulnerability as "the susceptibility to incidents that can result in considerable reductions in road network serviceability" has also been widely used since then (Berdica 2002). The literature on vulnerability indices can be broadly grouped into the distance-based and the cost-based approaches. Indices based on travel distance are relevant to sparse regional networks where drivers may need longer detours to reach their destinations if a link is blocked. On the contrary, several alternative routes are usually available in dense urban networks. Moreover, it is well known that drivers often prefer quicker routes that need not necessarily be shorter in distance terms. Hence, indices such as Hansen's Index (Taylor et al. 2006) and the efficiency measure (Latora and Marchiori 2001), which are based purely on distance, are considered unsuitable for measuring the vulnerability of dense urban road networks (Balijepalli and Oppong 2014).

The topic of network vulnerabilities goes beyond transportation planning and has its roots in graph theory and complex network research. The most common measure used to find vulnerabilities is the betweenness centrality of nodes or edges of the network, which measures the tendency of paths in a given network to pass through the element. This parameter implies that nodes or edges with high betweenness centrality tend to concentrate on more trips and be more critical in a network (Barthélemy 2004).

Spatiotemporal graph convolutional networks (STGCNs) are widely used in traffic prediction tasks because they model both temporal patterns and spatial dependencies across road networks. In more recent approaches, the STGCN models aim to make predictions of road segments' flooding status based on their own and adjacent road segments' current flooding status. These studies utilized high-resolution traffic data to infer the inundation status of road segments (road segments with no traffic speed were shown to be inundated). Hence, traffic speed data during floods can be a reliable indicator of road network status during floods, and using models for traffic flow predictions can be adjusted to derive foresight about road network flooding status (Yu et al. 2018).

Another study has emphasized the cascading effects of flooding on urban accessibility and emergency response. Yin et al. (2017) evaluated how sea-level rise and coastal flooding reduce emergency response access in Lower Manhattan using high-resolution flood maps and accessibility modeling. Their study highlighted the need for spatially detailed and temporally aware models in emergency planning. Similarly, Tahvildari et al. (2021) analyzed compound flooding scenarios and their impact on emergency response operations, incorporating hydrologic and hydraulic modeling to support planning decisions.

Pregnolato et al. (2017) developed a widely cited "depth-disruption function" that models the relationship between floodwater depth and vehicle speed, providing a critical improvement over binary open/closed road assumptions commonly used in flood impact models. Their empirically derived function enabled integration into transport simulations to estimate travel delays more accurately under varying flood depths. Fan et al. (2020) proposed a network percolation–based contagion model that captures the spatial and temporal evolution of flooding in urban road networks by treating flood spread analogously to epidemic transmission using an susceptible-exposed-infected-recovered (SEIR)-based mathematical framework. The model predicts the likelihood of road segments transitioning from functional to flooded states based on the status of neighboring segments and macroscopic contagion parameters.

From recent studies that evaluated roadway vulnerability, Marian et al. (2024) proposed a simulation-based framework to evaluate the vulnerability and robustness of urban road networks during flood-induced closures by analyzing both controlled intersections and uncontrolled freeway segments. Redzuan et al. (2022) developed a GIS-based methodology to evaluate the vulnerability of road segments based on the availability and efficiency of diversion routes. Their approach focused on constructing two disjointed alternative paths for each disrupted link, emphasizing connectivity recovery and accessibility under disruption scenarios.

Although these studies provided valuable insight in traffic flow estimates (Pregnolato et al. 2017) and an efficient way of forecasting flood propagation (Yin et al. 2017; Fan et al. 2020; Yuan et al. 2022), they do not support predictive, network-scale assessment nor do they incorporate indirect disruptions from noninundated segments such as increased congestion or speed reductions resulting from rerouted traffic. Also, some recent studies, like Marian et al. (2024), relied on predefined closure scenarios and mesoscopic simulation, which can be computationally intensive and lack predictive analytics.

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Additionally, current models lack the ability to forecast near-term roadway serviceability by integrating multisource data—including flood simulation, historical traffic patterns, and roadway characteristics—into a cohesive, predictive framework. There is also a lack of visual, interactive decision-support systems that translate these predictions into actionable insights. This study addresses these gaps by proposing a predictive artificial intelligence (AI)-enhanced digital twin framework that enables near-real-time serviceability assessment, considering both direct and indirect impacts of flooding on urban roadway networks.

Digital Twinning Visualization of Flood Impacts

Early research by Kuser Olsen et al. (2018) explored the use of DTs for flood risk communication through realistic interactive visualization techniques. Their study developed an integrative flood-risk map superimposed on a satellite base map via systems based on GIS, aiming to enhance public awareness and motivate proactive flood mitigation efforts. A comparative analysis with Hazard-US (HAZUS) system, published by FEMA (2011), demonstrated that although both approaches effectively conveyed flood risk knowledge, the DT-based system was superior in fostering engagement and motivating action. Despite lacking a full virtual model of physical infrastructure, this study illustrated the potential of DT visualization for emotionally impactful flood risk communication.

Ghaith et al. (2021) proposed a city-scale DT framework incorporating real-time simulations of Calgary, Canada's power, transportation, water distribution, and river systems. By continuously integrating sensory, Internet of Things (IoT), global positioning systems (GPS), and satellite data, the DT model dynamically adjusted its flood impact assessments. Although their implementation effectively visualized flood risks, it remained constrained by a limited simulation boundary, highlighting the need for broader-scale validation.

The Earth System Digital Twin (ESDT), a collaborative effort between National Aeronautics and Space Administration (NASA) and Centre National d'Etudes Spatiales (CNES), further exemplifies large-scale DT flood modeling (Huang et al. 2022). Suquet et al. (2023) expanded upon ESDT by integrating hydrologic models and AI-enhanced remote sensing data to generate rapid flood predictions. Their four-step DT pipeline incorporated real-time water level data, synthetic aperture radar (SAR) imaging, numerical modeling, and geo-located social media analytics to assess flood risks comprehensively. Their system prioritized interoperability and large-scale modeling, and future advancements could focus on enhanced visualization techniques for policymakers and the public.

The digital twin framework developed in this study enables the integration of flood simulation results, real-time and historical traffic data, and vulnerability metrics into a unified virtual representation of the roadway network. Unlike conventional GIS-based dashboards, this platform incorporates predictive AI models (via a graph attention network) to dynamically estimate roadway serviceability as conditions evolve. The digital twin environment facilitates real-time visualization of accessible and inaccessible segments and supports scenario-based decision-making during flood events. Although the current implementation does not yet include automated feedback or control actions back to field sensors (i.e., bidirectional flow), it supports forward simulation with near-real-time data feeds, aligning with operational definitions of digital twins in transportation resilience applications.

Methodology

This study proposes a way to define roadway serviceability and develop a new prediction method for the serviceability of flood-vulnerable roadways.

Definition of Roadway Serviceability for Disaster Resilience

In this study, roadway serviceability is defined as the ability of a roadway to consistently maintain minimum quality and performance of its structure and pavement and effectively and safely support its operations as its intended purpose. The ability encompasses the road's structural integrity, surface condition, and overall functionality, ensuring vehicles can travel without significant risk or difficulty. In extreme weather events such as flooding, roadway serviceability also involves assessing whether a road is passable, free from hazardous conditions, and suitable for emergency and regular traffic.

Although roadway serviceability can be defined broadly to include structural integrity, user comfort, and safety, in the context of short-term flood response, it is operationalized in this study using traffic speed as a practical and measurable proxy. Traffic speed reflects both direct disruptions (e.g., inundated or closed roadways) and indirect network effects (e.g., rerouting and congestion), making it a meaningful indicator of performance during emergencies. Although structural vulnerability is not modeled, flood exposure and roadway connectivity characteristics are included as node and edge features, enabling the model to predict segment-level serviceability under evolving flood conditions.

With this scope of the definition, the authors identified the relevant factors of roadway serviceability and established a fundamental baseline to calculate and predict roadway serviceability under catastrophic flooding events.

Prediction Method for the Serviceability of Flood-Vulnerable Roadways

Diverse data sources and streams can be used to assess the impact and damage of flooding on roadway systems. Among them, this research project utilizes historical traffic data recorded in Houston city to identify underlying patterns of traffic changes under flooding impacts and reveal the impacted sections of roadways. The proposed framework uses spatial data, i.e., sensor locations and their position compared with flood zones, and temporal data, that is, time intervals of every 5 min for speed data and every 6 h for flood data. Based on these data, the authors built a serviceability assessment and prediction method. In addition, the proposed method acts as a digital twin during flood events with real-time traffic amount data that first simulates the flood and traffic speed in a city by identifying the flooded zones and blocked roadways and second, predicts the near future condition of the flood and traffic based on a given time window. Fig. 1 shows the overall process and data flow of the study.

The first step of this project involves extracting and utilizing data obtained from previous research efforts, including flood simulations, identification of inundated roadways, and inundation durations. These data provide a foundation for subsequent serviceability and accurate traffic predictions based on the new conditions.

We selected a GAT over other GNN variants such as graph convolutional network (GCN) and GraphSAGE due to its ability to assign dynamic, learnable attention weights to neighboring nodes. This is especially valuable in flood-impacted road networks, where the influence of surrounding roadway segments on a given node can vary significantly depending on inundation levels and network topology. Unlike GCN, which treats all neighbors equally, GAT enables the model to learn which connections are more informative in degraded traffic conditions. Additionally, GAT has shown improved performance in prior work on spatiotemporal traffic forecasting due to its capacity to capture heterogeneous spatial dependencies.

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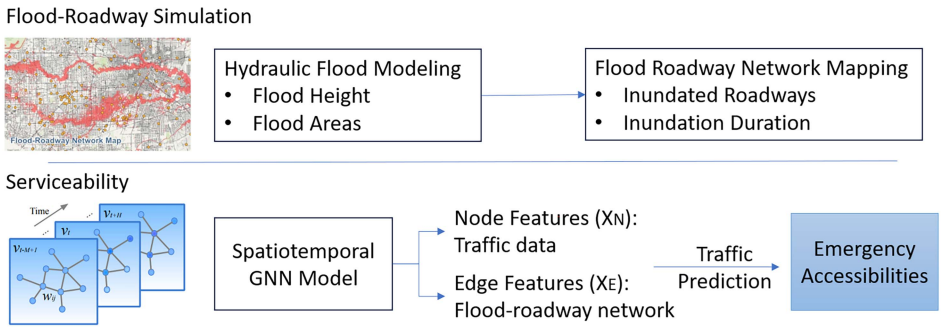


Fig. 1. Overall methodology. (Map by author using ArcGIS © Esri.)

Table 1. Node features for each time interval

Date	Time interval	Interval No.	Node features August 25	Node features August 26	Node features August 27
August 25	00:00–06:00	1	August 25 speed data	—	—
August 25	06:00–12:00	2	August 25 speed data	—	—
August 25	12:00–18:00	3	August 25 speed data	—	—
August 25	18:00–00:00	4	August 25 speed data	—	—
August 26	00:00–06:00	5	August 25 speed data	August 26 speed data	—
August 26	06:00–12:00	6	August 25 speed data	August 26 speed data	—
August 26	12:00–18:00	7	August 25 speed data	August 26 speed data	—
August 26	18:00–00:00	8	August 25 speed data	August 26 speed data	—
August 27	00:00–06:00	9	August 25 speed data	August 26 speed data	August 27 speed data

Table 2. Edge features for every time interval

Date	Time interval	Interval no.	Edge feature	Edge feature	Edge feature	Edge feature	Edge feature	Edge feature
August 25th	00:00–06:00	1	Distance	Flood data1	—	—	—	—
August 25th	06:00–12:00	2	Distance	Flood data1	Flood data2	—	—	—
August 25th	12:00–18:00	3	Distance	Flood data1	Flood data2	Flood data3	—	—
August 25th	18:00–00:00	4	Distance	Flood data1	Flood data2	Flood data3	Flood data4	—
August 26th	00:00–06:00	5	Distance	Flood data1	Flood data2	Flood data3	Flood data4	Flood data5

The GAT model developed by Zhang et al. (2019), which is a type of the GNN model, is used for predicting the traffic given the speed data every 5 min (Zhang et al. 2019). This research team attained the accumulated traffic data from INRIX, a database of vehicles' average speed recorded every 5 min in Houston city during Hurricane Harvey from August 25 to August 30, 2017.

The proposed method also incorporates flood data into the model and the distances between the nodes. In this GAT model, each node represents a sensor that contains information about the average speed of the vehicles every 5 min. The edges represent the connection between the nodes that are based on two factors: (1) the distance between the nodes attained from the formula used in the Zhang et al. (2019) study, and (2) the existence of a road inundation between every two consecutive nodes based on the flood data extracted from the previously developed flood model (Yin et al. 2024). The model predicts the speed of the vehicles 45 min in the future based on the node and edge features' data for every 6-h time interval that the flood model updates.

The time intervals of the flood prediction model are every 6 h starting from August 25 at midnight (00:00); for example, the first interval is August 25 from 00:00 to 06:00, and the next is from 06:00. to 12:00 the same day. Node features are speed data for every sensor (node), and the edge features are (1) the distance between the nodes, and (2) the flood data for each time interval.

For every time interval, an edge feature of flood data from that interval was added to the nodes, and the traffic prediction model was executed separately. The edge feature shows the properties of the connections between two nodes. For instance, the distance edge feature helps identify the neighboring sensors, and the flood edge feature helps the GAT model identify the changes in connection between the nodes due to flood inundation.

As an example, at the first-time interval (August 25 from 00:00 to 06:00), the node features are speed data, and the edge features are the distance between the nodes and the flood data about the first interval, i.e., the roadways that were flooded from 00:00 to 06:00. For the second interval, the node feature is again the speed data, and the edge features are distances between the nodes, flood data at the first interval, and flood data at the second interval. This process continues for every step until all intervals are added to the prediction model. In summary, for every time window, an edge feature is added to the model, along with the speed data added as node features. Table 1 presents the node features for every time interval, and Table 2 provides a part of the edge features for each time interval.

Flood data are the information extracted from the flood model in the Yin et al. (2024), which determines the areas that were flooded during each time interval with specific depth. The points in Fig. 2 represent the location of each sensor of an area in Houston and their position relative to them. This study selected the roadway sections (between each of two

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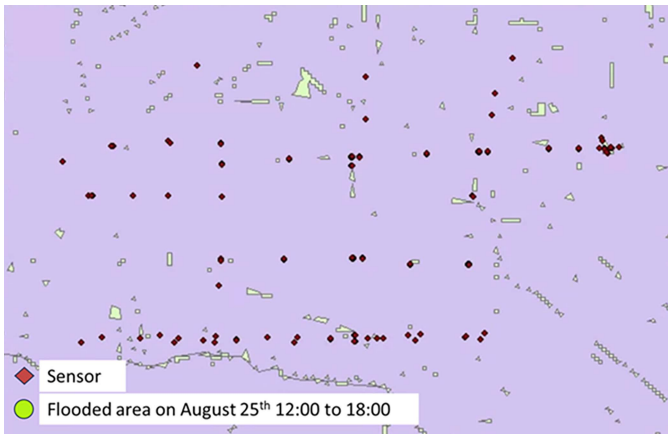


Fig. 2. Nodes (sensors) that contain the speed data as node feature values.

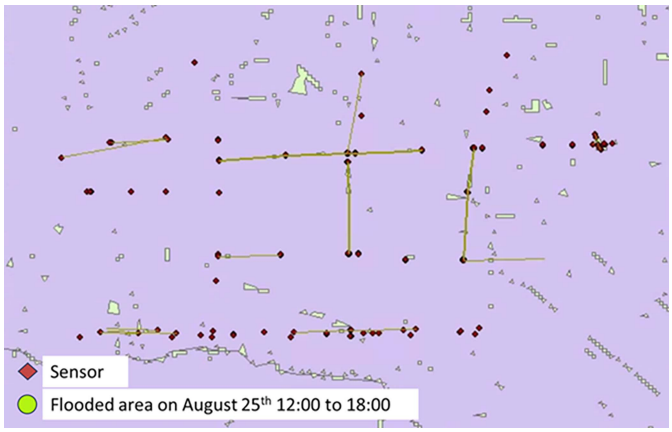


Fig. 4. Distinguishing the lines that intersect with the flooded areas and using them as edge features in the GAT model. Time interval 3 is used as an example.

consecutive sensors) with flood depths higher than 58.4 cm (23 in.). It was assumed that this flood depth will stop most vehicles from passing, and there should be a significant drop in the speed data.

This was applied through the XY-to-Line feature in ArcMap software (version 10.8) by first adding a line between each consecutive node (i.e., connecting the closest sensors by drawing a line between them (Fig. 3) and then selecting the lines and nodes that intersect the flooded area (Fig. 4) for each time interval. Fig. 3 illustrates the XY-to-Line feature for all nodes. Fig. 4 shows an example of this selection method in the third time interval, on August 25 from 12:00 to 18:00. The lines intersecting with the flooded areas are distinguished. Fig. 2 shows the nodes that contain the speed data as node features, and Figs. 3–5 are the process of selecting the edge features based on flood data extracted from the flood model.

As an illustrative example, Fig. 5 shows the process of identifying the inundated roadway sections. Fig. 5(a) shows the points, and each point represents a sensor in a specific location. According to the flood model, the arrow in Fig. 5(b) depicts an inundated area. In Fig. 5(c), a line is drawn as a specification of an inundated roadway section, which will later be added to the GAT model as an edge feature for flood data.

The edge data in the model are a binary type in which if the edge (the connection between the nodes) is flooded, then it is equal to one. Otherwise, it is equal to zero. The lines in Fig. 3 represent the edges between the two sensors (nodes) where the roadway is flooded. This means that if the roadway between two sensors is inundated with floods of over 58.4 cm (the maximum flood depth allowing emergency service vehicles to operate), the edge is selected, and the flood data for that edge in the specific time interval are equal to one. It is expected that, at least in this section, the speed data will decrease significantly. It is important to note that the flood model's accuracy affects the prediction model's accuracy. Table 2 presents the edge features added for each time interval based on flood data extracted from the flood model using this method.

Fig. 6 depicts part of the process of adding the flood data as edge features for each round of executing the GAT model. This process continues until midnight on August 30 (based on the time that the flood water receded).

It is expected that the model cannot provide an accurate prediction for the first few intervals because the speed data and flood data are not adequate. However, by adding the flood data as an edge feature, the model's accuracy is expected to increase. Mean square error (MSE) was selected as the loss function to estimate the model's accuracy for each execution in which the new node and edge features are added. For each interval, as the flood data are added to the edge features, the MSE slightly decreased, which shows a slight increase in the GAT model's accuracy for each iteration. However, after Interval 12, a slight increase is evident in the MSE results for each interval, which is explained in the "Discussion" section.

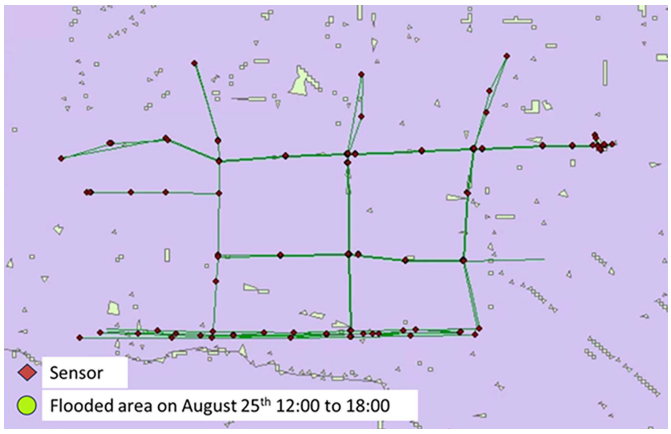


Fig. 3. Connecting all the nodes using XY-to-Line feature in ArcMap.

Digital Twin–Based Implementation and Method Validation

Digital twins are virtual replicas of physical infrastructure that facilitate real-time monitoring, analysis, and predictive modeling. In this study, the digital twin concept was employed to implement, assess, and validate the accuracy, reliability, and applicability of the proposed serviceability framework. Two primary motivations guided the use of digital twin–based validation. First, the dynamic nature of digital twins enables real-time interactions and continuous updating of roadway conditions in response to evolving flood scenarios. Serviceability must be dynamically calculated and adopted to reflect the conditions of vulnerable and affected communities before, during, and after disaster events.

In addition, a digital twin allows for the integration of essential and supplementary parameters, such as hydrological simulations, traffic disruptions, and roadway vulnerability indices, within an urban infrastructure context. By encompassing multiple data streams into an interactive visualization environment, the digital twin enhances the practical applicability of the GAT model and offers an

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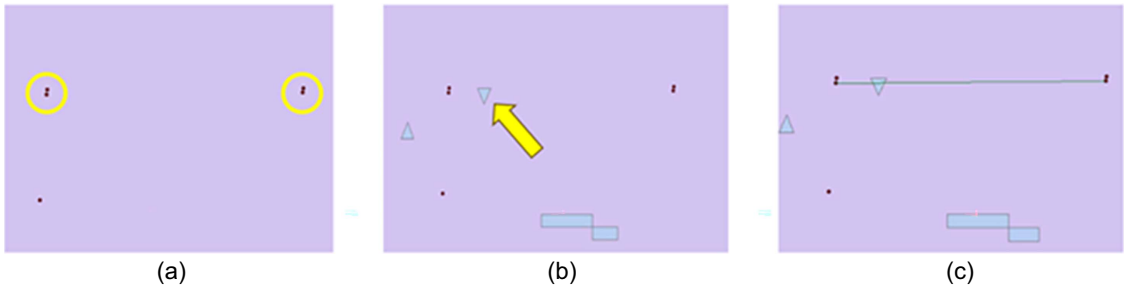


Fig. 5. Process of identifying inundations between two sensors.

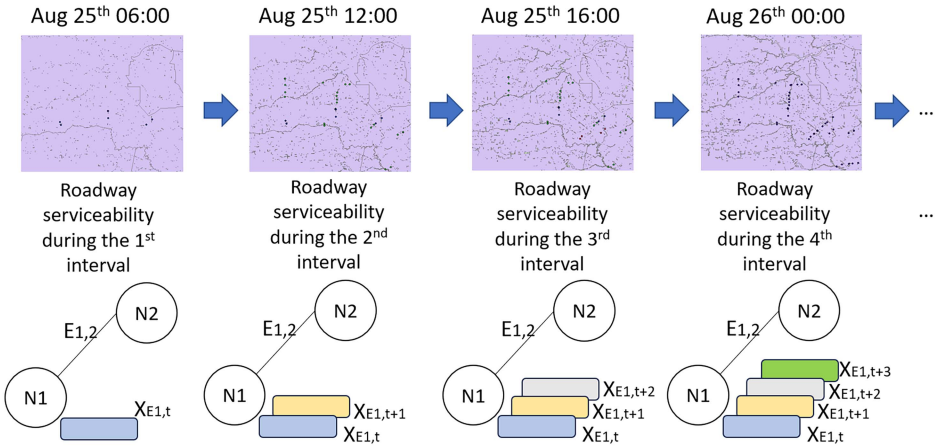


Fig. 6. Adding flood data as edge features for each time interval for the GAT model.

intuitive, actionable representation of roadway vulnerability under flood conditions.

The digital twin system is developed using GIS-based visualization techniques, integrating the following:

- Flood inundation maps generated using hydrological simulations (Yin et al. 2024).
- Traffic disruption data derived from real-time traffic sensors and historical patterns.
- Roadway vulnerability index calculated based on pavement deterioration factors and flood impact severity.

The visualization platform enables interactive analysis of flooding impacts, allowing users to intuitively visualize disaster scenarios and assess roadway accessibility under flood impact (Section "Validation with a Case Study").

Digital Twin–Based Visualization Framework

The digital twin–based visualization system is designed as a comprehensive framework for visualizing roadway serviceability prediction results in conjunction with flood-affected areas. This system enables experienced engineers to visually validate predictions, detect potential anomalies, and refine the assessment of roadway functionality under flood scenarios. Leveraging pre-existing flood simulation results, the system incorporates roadway inundation data as input to the GNN model to predict roadway serviceability. Operating through a multilayered architecture, this system ensures seamless data processing, model interoperability, and visualization for decision-making support (Fig. 7).

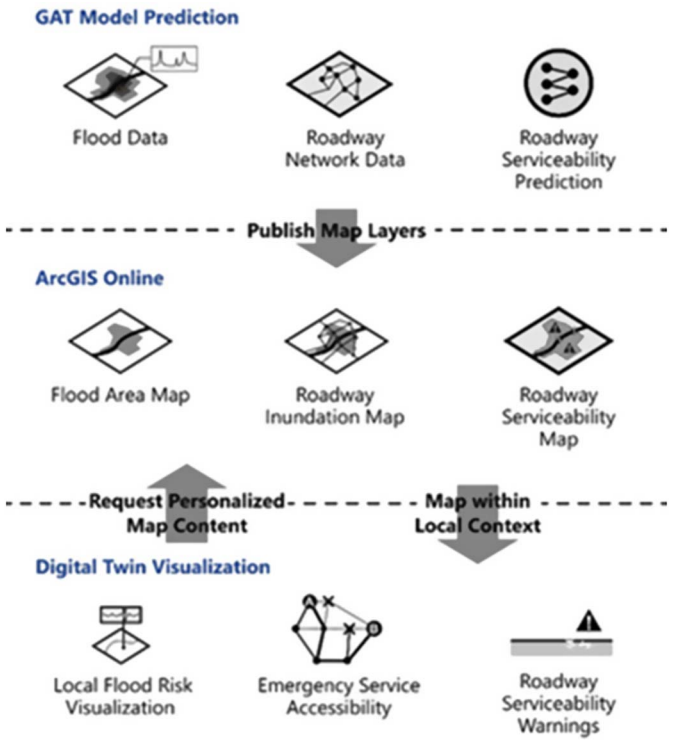


Fig. 7. Network architecture of the digital twin–based visualization framework.

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The digital twin–based visualization framework consists of three primary components that work in concert to provide accurate and real-time assessments of flood-related roadway serviceability.

The first component is the integrated GAT model prediction layer, which processes roadway inundation data derived from prior flood simulations. This layer employs the GAT model to predict roadway serviceability using roadway inundation data as input. Unlike traditional simulations, this layer focuses solely on predicting roadway serviceability based on the GAT model developed in this study, which is trained by leveraging historical INRIX traffic data and flood simulation results from studies conducted by Yin et al. (2024).

The second component is ArcGIS (version 11) Online, which functions as a middleware for publishing the GAT model prediction layer results online. This enables end-users across the internet to access and visualize prediction results on their devices. ArcGIS Online integrates predictive results from the GAT model with geographic information, allowing for seamless overlaying of flood maps, road networks, and serviceability predictions. This cloud-based platform facilitates data sharing, collaboration, and real-time access to visualization outputs for engineers and decision-makers.

The third component, the digital twin visualization layer, retrieves data from the cloud-stored maps in the second layer. This layer processes and interprets the stored spatial-temporal prediction data, transforming it into an interactive and real-time representation of roadway inundation and serviceability conditions. This layer converts the GAT model outputs stored in ArcGIS Online into an intuitive, immersive, and dynamic visualization environment where stakeholders can gain further insights based on human expertise. Therefore, the digital twin visualization enhances decision-making by offering a comprehensive view of affected areas within their urban context, enabling engineers to validate model predictions and refine assessments through interactive analysis.

The visualization framework integrated multiple data layers, including satellite-based base maps, three-dimensional (3D) terrain elevation data, and 3D building models obtained from public sources. INRIX roadway data and simulated flood extents (Yin et al. 2024) were incorporated as vector tile layers to represent dynamic traffic and flooding conditions. Finally, flood–roadway speed predictions developed in this study were integrated as time-series point data. Together, these layers provided the spatial and temporal foundation for the digital twin environment.

At this stage of development, the system functions as a digital twin prototype. Although the platform integrates precomputed flood simulation outputs and historical traffic data for predictive modeling, real-time ingestion of live traffic sensor data is not yet automated. The flood scenarios used in this version are not dynamically generated on demand but selected from a library of historical events. However, the input information of the model can be fed from sensors in real-time. Furthermore, the computational time can be reduced to a level that supports near-real-time simulation output. These limitations are acknowledged as current constraints of the operational implementation. Nevertheless, the system architecture supports future expansion to include live sensor feeds, automated model updates, and potential two-way feedback mechanisms to qualify as a fully operational digital twin.

Validation with a Case Study

To demonstrate the applicability of the proposed framework, a case study was conducted using the Hurricane Harvey flooding scenario. Although a real-time sensor data acquisition system was not established, the study employed reliable mock-up data sets derived from the simulation and historical flood conditions. These data sets were used to emulate real-time inputs, enabling the testing and illustration of the model's performance in predicting roadway serviceability during large-scale flood events.

Case Study

The objective of this case study is to evaluate the model's ability to predict roadway speed data and illustrate its practical utility in identifying road segments likely to remain serviceable in the immediate future, specifically, over the next 45 min. The model provides real-time speed data for each roadway section and predicts future speed conditions by incorporating information on roadway inundation, which is extracted from a flood model.

Unlike conventional approaches that focus only on flooded road sections, this model also predicts speed changes on noninundated roadways. These roadways may experience congestion due to traffic rerouted from blocked or submerged sections, impacting their serviceability. By considering both direct and indirect flood effects, the model provides a more comprehensive assessment of roadway conditions during extreme weather events.

For instance, during disasters like Hurricane Harvey, first responders can leverage the model's speed predictions to identify the fastest available routes to reach critical destinations efficiently. This forward-looking functionality enhances emergency response efforts by allowing responders to proactively navigate dynamic and often unpredictable roadway conditions with improved situational awareness and confidence.

Implementation and Assessment

The proposed model can be implemented as a digital twin to provide real-time insights into roadway serviceability during flood events by continuously forecasting roadway speeds based on evolving flood conditions. This digital twin is regularly updated with sensor data,simulation outputs, and predictive analytics. Fig. 8 demonstrates a real-world application of the digital twin approach, depicting a flood-predicted area in Houston during Hurricane Harvey on August 26, 2017, at 06:00. The flooded roadways and their corresponding predicted conditions are mapped onto a 3D visualization, where highlighted areas indicate inundation and boxes display current and predicted speeds for specific roadway segments. Each roadway segment is identified by its unique ID, allowing for precise monitoring and prediction updates at 45-min intervals, which is critical for tracking accessibility during disaster response.

Fig. 9 illustrates how the digital twin model dynamically tracks roadway serviceability over time by integrating flood predictions and real-time traffic data. In this example, the flood model forecasts inundation for August 28, 2017, at 18:00, and the speed data shown correspond to 18:45, demonstrating the evolving impact of flooding on roadway accessibility.

Compared with earlier conditions shown in Fig. 8, several roadway segments that were previously serviceable have become nonfunctional due to direct inundation or traffic disruptions caused by adjacent flooded roads. For example, Segment 1595369732, which had a recorded speed of approximately 43 km/h (27 mi/h) 2 days earlier, now exhibited a predicted speed of 0 km/h. This drastic change suggests complete inaccessibility, whether from water coverage or extreme traffic congestion resulting from rerouted vehicles.

The digital twin continuously updates these predictions every 45 min, synchronized with flood model forecasts updated every 6 h. This temporal resolution enables near-real-time monitoring of roadway conditions, enhancing situational awareness for emergency responders and transportation agencies because it helps them identify alternative routes, anticipate blockages, and allocate resources

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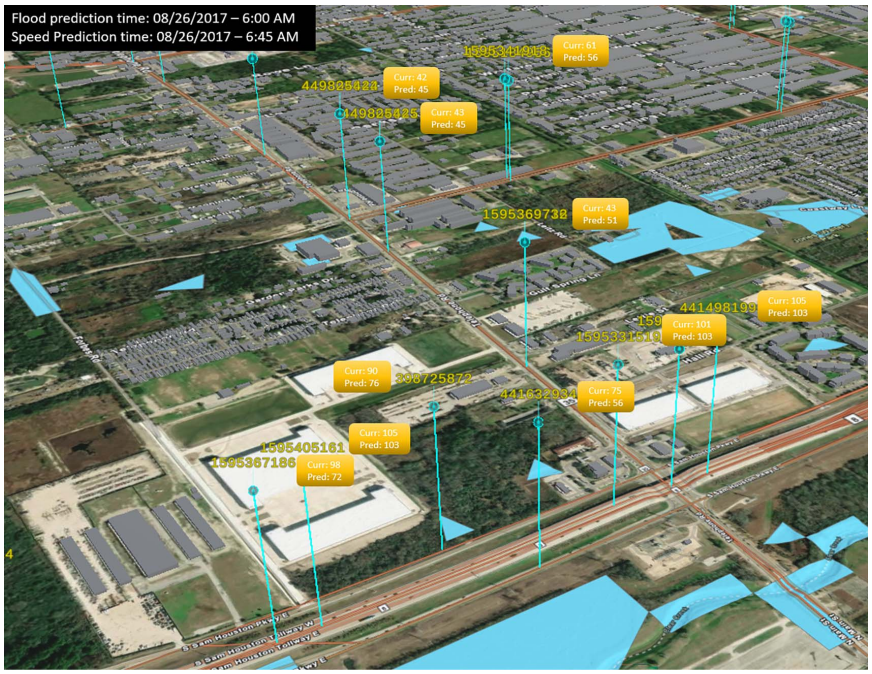


Fig. 8. Flood map prediction on August 26, 2017, of a selected area and speed prediction of each roadway segment. (Map from ArcGIS Terrain3D, Sources: Airbus DS, USGS, NGA, NASA, CGIAR, GEBCO, N Robinson, NCEAS, NLS, OS, NMA, Geodatastyrelsen and the GIS User Community, © OpenStreetMap contributors and the GIS User Community.)

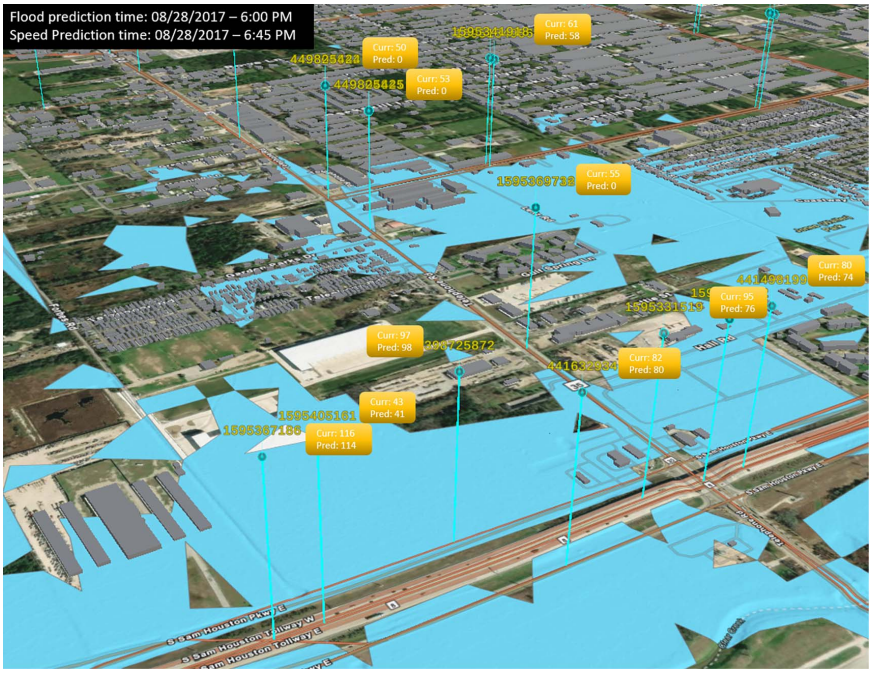


Fig. 9. Flood map prediction on August 28, 2017, of the same selected area in Fig. 8, and speed prediction of each roadway segment. (Map from ArcGIS Terrain3D, Sources: Airbus DS, USGS, NGA, NASA, CGIAR, GEBCO, N Robinson, NCEAS, NLS, OS, NMA, Geodatastyrelsen and the GIS User Community, © OpenStreetMap contributors and the GIS User Community.)

more efficiently. By visualizing roadway serviceability holistically, the model ensures that decision-makers are not only aware of flooded segments but also of secondary impacts on noninundated roadways affected by shifting traffic patterns.

This example underscores the capacity of the model to provide a real-time, adaptive representation of roadway functionality in dynamic flood scenarios. As such, it serves as a valuable decision-support tool for disaster response, traffic management, and infrastructure resilience planning.

Model Training and Evaluation

The GAT model was trained using INRIX traffic speed data for the Houston metropolitan area during Hurricane Harvey

(August 25–30, 2017). The model predicts future traffic speed conditions on roadway segments using historical traffic patterns and spatial flood intensity as input features.

Each road segment is represented as a node in the graph, with edges weighted based on physical proximity. We used a distance-based exponential weighting function with $\sigma^2 = 0.1$ and $\varepsilon = 0.5$ to promote sparsity. The "USE_GAT_WEIGHTS" flag was set to "True", which binarizes the adjacency matrix and includes self-loops, following the original GAT formulation. Edge attributes include a 12-dimensional time series of flood exposure levels across multiple time stamps.

The data set was constructed using a sliding time window where each graph snapshot included 12 historical time steps (x) and 9 prediction time steps (y), covering the 45-min future speed prediction. Data were standardized using z-score normalization. We split the data set using Temporal_Signal_Split with an 80:20 time-based ratio, ensuring temporal continuity between training and testing sets.

Model training was conducted over 15 epochs using minibatches of 2,000 graph snapshots per epoch. Training was performed on a CPU device using the Adam optimizer with a learning rate of 0.01. The model optimized MSE between predicted and observed traffic speeds. Evaluation was conducted on a held-out 20% time-based test split, where the lowest test MSE was 0.2602. This demonstrates the model's ability to generalize to new conditions and accurately estimate short-term traffic performance during flood events. In the next section, we further discuss variations in MES scores attained from the model in different intervals.

Methodological Contributions and Novelty

This study presents several key methodological contributions that extend the current state of the art in flood-related serviceability modeling:

- Coupling of hydrodynamic simulation with GAT-based prediction: Unlike prior models that treat flood effects in isolation or use fixed closure scenarios, this framework dynamically fuses flood simulation outputs with real-time and historical traffic data using a GAT. This coupling enables predictive modeling of both direct and indirect roadway disruptions.
- Serviceability estimation at segment level: Whereas most prior works predicted binary status (flooded versus unflooded), the model proposed in this study forecasts a continuous serviceability score—capturing varying degrees of speed degradation and accessibility at the road segment level, including for segments not directly inundated but indirectly impacted by rerouting and congestion.
- Use of GAT architecture for indirect impact modeling: Our implementation leverages the GAT's attention mechanism to dynamically weigh the influence of neighboring segments, allowing the model to capture localized traffic propagation effects during floods—something traditional GNNs or statistical methods cannot do.
- Integration with a GIS-based digital twin: The predicted outputs are embedded within an operational digital twin platform that supports real-time visualization, scenario exploration, and emergency decision support, bridging the gap between analytical models and end-user deployment.

These innovations enable more accurate, spatially aware, and operationally actionable insights compared with prior flood disruption models that focused only on inundation status or used rigid traffic closure assumptions.

Results and Discussion

The proposed method is developed based on real traffic and flooding simulation data in conjunction with graph neural network models to assess roadway serviceability from a short-term perspective. It considers two critical factors: roadway inundation and the resulting traffic.

The accuracy of the prediction model is influenced by the following key elements:

- The predictive accuracy of the flood model, which directly affects the overall accuracy of the prediction for the serviceability of the roadways.
- The volume and granularity of traffic data used for training and testing the machine learning algorithm impacting the model's learning and generalization capabilities.

These interdependent components collectively shape the effectiveness of the proposed method in forecasting roadway accessibility during and after extreme flooding events.

To evaluate the performance of the GAT model, this study used the MSE metric, which assesses the average squared difference between the actual and predicted values. The results of the MSE show that the inclusion of flood model data as edge data in the GAT model contributed to improved predictive accuracy of the model. Fig. 10

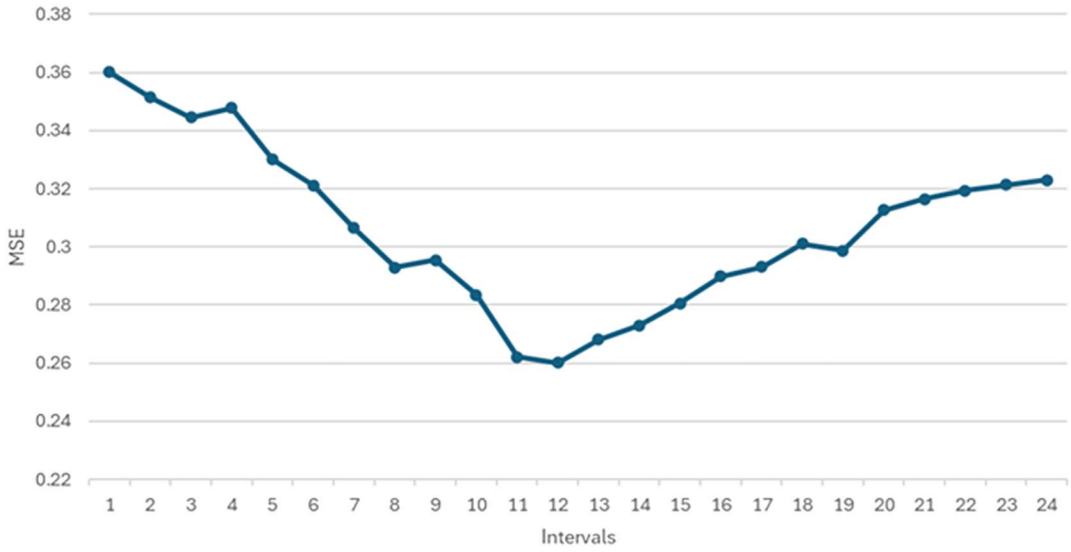


Fig. 10. Variations in MSE for each interval.

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shows the variations in MSE for each time interval. After the 12th interval, however, the MSE increased, which shows a decrease in the model's accuracy.

A primary factor contributing to this decline is that this study focuses on predicting flooded areas and inundated roadways, whereas the duration of inundation remains a subject for future study. Although the flood model successfully identified the locations of flooding, the actual recession of floodwaters occurred sooner than predicted. This discrepancy may be attributed to mitigation efforts by authorities, such as drainage operations and other flood management measures, which were not explicitly accounted for in the model due to the lack of accurate and detailed input data sets. In addition, to accurately evaluate the impact of flooding, dynamic modeling and simulation are required, representing changing parts and portions of roadways affected by inundation over time. This study, however, employed static flooding data that illustrates inundated areas of Houston city at specific time intervals (6 h).

Despite the promising results, the study encountered some limitations related to the volume and scope of traffic data in Houston for model training and testing. Due to computational constraints and data handling capacity, only two zones in the Houston area—selected based on zip codes—were analyzed. Expanding the model to incorporate broader geographic coverage and more comprehensive data sets is expected to significantly enhance prediction accuracy. In addition, the model does not account for other factors, such as a temporary event, including a car accident, that can make traffic changes that are not affected by a flooding disaster. These factors introduce uncertainty into the relationship between flooding and traffic behavior, limiting the granularity of analysis.

Another important factor influencing the accuracy of the prediction model is the spatial and temporal resolution of the input data. Higher resolution flood and traffic data can provide more precise insights into roadway conditions, allowing for finer-grained predictions of serviceability. Additionally, integrating real-time sensor data, such as traffic cameras and GPS-based vehicle tracking, could enhance the model's responsiveness to rapidly changing conditions.

Even though the current case study was limited to two zones in Houston due to computational constraints and data-handling complexity, the framework is designed to be generalizable to other regions. Its modular structure allows for the integration of location-specific data such as flood exposure, roadway topology, and historical speed observations. To apply this approach in different contexts (e.g., coastal cities with storm surge, or mountainous areas prone to flash flooding), appropriate adjustments to the flood modeling inputs and network features would be required. As long as adequate traffic, roadway-segment, and flood data are available, the underlying GNN-based architecture and inference pipeline can be retrained to accommodate regional characteristics. Future work will explore scalability and domain adaptation to enable broader geographic application.

The digital twin visualization offers a more intuitive representation of roadway vulnerabilities, facilitating proactive disaster response and infrastructure planning. The ability to fully visualize multiple flood scenarios in an interactive, three-dimensional, and geospatially accurate way aids transportation agencies in better assessing flood impacts in real time. This enhanced situational awareness facilitates the optimization of evacuation routes, prioritization of infrastructure repairs, and mitigation of long-term roadway degradation. By offering timely, actionable insights, the digital twin serves as a critical tool for improving urban resilience in flood-prone regions.

Although this framework adopts key principles of a digital twin—data integration, predictive modeling, and interactive visualization—it is currently limited in terms of automated real-time data acquisition and dynamic model execution. Live traffic feeds were used for training and validation, but are not yet ingested in real time. Similarly, flood scenarios were simulated offline and selected during analysis rather than being generated dynamically within the interface. As future work, we plan to establish automated data pipelines that can ingest traffic and flood sensor data in real time, and to expand the model's feedback capabilities to allow real-time reconfiguration and alert generation. These enhancements will strengthen the system's alignment with full digital twin functionality.

Although the physical sensor network (i.e., node positions and distances) is fixed, the graph structure in this study is dynamic because edges are defined based on the presence of flood inundation between sensor nodes. Specifically, an edge exists between two nodes only if the flood model indicates active inundation along the path connecting them. As floodwaters rise or recede, these connections are dynamically updated—edges are created when inundation appears and removed when it subsides—effectively reconfiguring the graph to reflect evolving roadway accessibility. This data-driven structure allows the model to capture both the direct effects of flooding (blocked or inaccessible roadways) and indirect consequences (e.g., rerouting or congestion) through predicted speed data. Although the model does not simulate vehicle-level trajectories, its dynamic edge representation provides a robust foundation for short-term, networkwide impact assessment during flood events.

This study does not include a calibrated long-term pavement deterioration model. The VF2 factor (mentioned in the "Literature Review" section) was used as a static input derived from prior knowledge and infrastructure records. Future work may expand the framework to simulate multievent pavement degradation and recovery processes.

Conclusion

Securing operational and serviceable roadways under diverse disaster scenarios is always imperative to maintain public safety, evacuation routes, and resiliency. In addition, its prediction and continuing maintenance are equally critical for the timely delivery of emergency supplies, disaster relief, and long-term recovery operations. As such, assessing roadway serviceability during flood disasters is essential to enhance the safety of our neighborhood and improve long-term disaster resilience. Although serviceability plays a pivotal role in managing disaster preparedness and response, it will be critical in promoting roadway structure and pavement design and proposing necessary changes in codes and materials. In this regard, this study proposes a new perspective regarding graph neural networks–based serviceability evaluation and prediction.

Moreover, this research established a critical baseline for creating a digital twin capable of modeling and evaluating the multifaceted impact of flooding on roadway serviceability and vulnerability at a network level. Because the digital twin entails a robust, comprehensive data hub that can connect multiple sensor-based data streams and real-time captured information from the physical twin, serviceability-based data analyses and disaster-related prediction simulations will be more promoted with the diverse data layers.

This study leverages variations in traffic speed and offers a reliable method for forecasting road serviceability during flooding and the availability of road segments shortly, within the next 2–6 h, during flood events. The study provided the following key research contributions:

- Introduction of a formal definition of short-term roadway serviceability under flood conditions that incorporates both physical vulnerability and functional operability (including traffic rerouting effects).

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- Developent of a hybrid AI-hydrodynamic framework that couples flood simulation with a graph attention network to forecast speed and accessibility across a road network in near real time.
- Demonstration that indirect impacts on nonflooded segments can be captured using a graph-based attention mechanism, advancing current flood-response models beyond binary disruption classifications.
- Validation of the predictive capability of the model using real traffic and flood data from Hurricane Harvey, highlighting the system's value in time-sensitive emergency response contexts.

To accurately assess the serviceability of roadways under diverse flooding scenarios, a variety of factors and disaster aspects, including the current roadway conditions, should be considered. This study primarily utilized traffic and flooding data to explore the underlying interrelationships between roadway inundation and traffic behavior, particularly the effect of flood events on traffic volume and flow. Given the availability of historical data sets from previous disasters, applying the approach to the Houston area yielded reliable insights into serviceability prediction and analysis.

However, to enhance the model's applicability to broader and more vulnerable regions, future efforts must incorporate additional serviceability-related variables into the proposed methodology. With the identified relationships between traffic data and roadway inundation, we can evaluate flood vulnerability and pavement deterioration, including surface distress. In addition, the outcomes of this study will be vital for assessing future traffic flow and access disruptions, such as road closures and detour plans during flooding events. In addition, the knowledge can be employed to determine roadway resilience and adaptation strategies according to their damage levels and long-term durability. By incorporating flooding considerations into roadway serviceability analysis, engineers and planners can better design, maintain, and improve roads to ensure they remain functional and safe even in flood-prone regions.

Looking forward, the study opens multiple avenues for advancing resilience strategies in transportation. Incorporating higher-resolution flood modeling, refining serviceability definitions using real-time pavement condition monitoring, and expanding the geographic applicability of the framework will further enhance its utility. Furthermore, coupling this model with climate adaptation planning and equity-focused mobility assessments could ensure more inclusive and forward-looking urban resilience initiatives. As climate-induced extreme weather events continue to escalate, the integration of AI, digital twins, and dynamic infrastructure modeling will become indispensable tools for safeguarding transportation systems and the communities they serve.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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Author Contributions

Moeid Shariatfar: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Validation; Visualization; Writing – original draft; Writing – review and editing. Yangtian Yin: Data curation; Formal analysis; Validation; Visualization; Writing – original draft; Writing – review and editing. Yong-Cheol Lee: Conceptualization; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Supervision; Validation; Visualization; Writing – original draft; Writing – review and editing. Kunhee Choi: Funding acquisition; Investigation; Supervision; Writing – original draft; Writing – review and editing.

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